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LearnCpp.com is a free website devoted to teaching you how to program in modern C++. The lessons on this site will walk you through all the steps needed to write, compile, and debug your C++ programs. No prior programming experience is necessary, but programmers of all levels will benefit from our best practices, tips, and insights.

Becoming an expert won't happen overnight, but with a bit of patience, you'll get there. And LearnCpp.com show you the way.

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Introduction / Getting Started

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- 0.1 [Introduction to these tutorials](https://www.learncpp.com/cpp-tutorial/introduction-to-these-tutorials/)
- 0.2 [Introduction to programs and programming languages](https://www.learncpp.com/cpp-tutorial/introduction-to-programming-languages/)
- 0.3 [Introduction to C/C++](https://www.learncpp.com/cpp-tutorial/introduction-to-cplusplus/)
- 0.4 [Introduction to C++ development](https://www.learncpp.com/cpp-tutorial/introduction-to-cpp-development/)
- 0.5 [Introduction to the compiler, linker, and libraries](https://www.learncpp.com/cpp-tutorial/introduction-to-the-compiler-linker-and-libraries/)
- 0.6 [Installing an Integrated Development Environment \(IDE\)](https://www.learncpp.com/cpp-tutorial/installing-an-integrated-development-environment-ide/)
- 0.7 [Compiling your first program](#)

[\(https://www.learncpp.com/cpp-tutorial/compiling-your-first-program/\)](https://www.learncpp.com/cpp-tutorial/compiling-your-first-program/)

0.8 A few common C++ problems

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0.9 Configuring your compiler: Build configurations

[\(https://www.learncpp.com/cpp-tutorial/configuring-your-compiler-build-configurations/\)](https://www.learncpp.com/cpp-tutorial/configuring-your-compiler-build-configurations/)

0.10 Configuring your compiler: Compiler extensions

[\(https://www.learncpp.com/cpp-tutorial/configuring-your-compiler-compiler-extensions/\)](https://www.learncpp.com/cpp-tutorial/configuring-your-compiler-compiler-extensions/)

0.11 Configuring your compiler: Warning and error levels

[\(https://www.learncpp.com/cpp-tutorial/configuring-your-compiler-warning-and-error-levels/\)](https://www.learncpp.com/cpp-tutorial/configuring-your-compiler-warning-and-error-levels/)

0.12 Configuring your compiler: Choosing a language standard

[\(https://www.learncpp.com/cpp-tutorial/configuring-your-compiler-choosing-a-language-standard/\)](https://www.learncpp.com/cpp-tutorial/configuring-your-compiler-choosing-a-language-standard/)

0.13 What language standard is my compiler using?

[\(https://www.learncpp.com/cpp-tutorial/what-language-standard-is-my-compiler-using/\)](https://www.learncpp.com/cpp-tutorial/what-language-standard-is-my-compiler-using/)

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C++ Basics

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1.1 Statements and the structure of a program

[\(https://www.learncpp.com/cpp-tutorial/statements-and-the-structure-of-a-program/\)](https://www.learncpp.com/cpp-tutorial/statements-and-the-structure-of-a-program/)

1.2 Comments [\(https://www.learncpp.com/cpp-tutorial/comments/\)](https://www.learncpp.com/cpp-tutorial/comments/)

1.3 Introduction to objects and variables

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1.4 Variable assignment and initialization

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1.5 Introduction to iostream: cout, cin, and endl

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[\(https://www.learncpp.com/cpp-tutorial/uninitialized-variables-and-undefined-behavior/\)](https://www.learncpp.com/cpp-tutorial/uninitialized-variables-and-undefined-behavior/)

1.7 Keywords and naming identifiers

[\(https://www.learncpp.com/cpp-tutorial/keywords-and-naming-identifiers/\)](https://www.learncpp.com/cpp-tutorial/keywords-and-naming-identifiers/)

1.8 Whitespace and basic formatting

[\(https://www.learncpp.com/cpp-tutorial/whitespace-and-basic-formatting/\)](https://www.learncpp.com/cpp-tutorial/whitespace-and-basic-formatting/)

1.9 Introduction to literals and operators

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1.10 Introduction to expressions [\(https://www.learncpp.com/cpp-tutorial/introduction-to-expressions/\)](https://www.learncpp.com/cpp-tutorial/introduction-to-expressions/)

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1.x Chapter 1 summary and quiz

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C++ Basics: Functions and Files

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- 2.1 Introduction to functions (<https://www.learncpp.com/cpp-tutorial/introduction-to-functions/>)
- 2.2 Function return values (value-returning functions)
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- 2.3 Void functions (non-value returning functions)
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- 2.4 Introduction to function parameters and arguments
(<https://www.learncpp.com/cpp-tutorial/introduction-to-function-parameters-and-arguments/>)
- 2.5 Introduction to local scope (<https://www.learncpp.com/cpp-tutorial/introduction-to-local-scope/>)
- 2.6 Why functions are useful, and how to use them effectively
(<https://www.learncpp.com/cpp-tutorial/why-functions-are-useful-and-how-to-use-them-effectively/>)
- 2.7 Forward declarations and definitions
(<https://www.learncpp.com/cpp-tutorial/forward-declarations/>)
- 2.8 Programs with multiple code files
(<https://www.learncpp.com/cpp-tutorial/programs-with-multiple-code-files/>)
- 2.9 Naming collisions and an introduction to namespaces
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- 2.10 Introduction to the preprocessor
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- 2.11 Header files (<https://www.learncpp.com/cpp-tutorial/header-files/>)
- 2.12 Header guards (<https://www.learncpp.com/cpp-tutorial/header-guards/>)
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Debugging C++ Programs

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- 3.3 A strategy for debugging (<https://www.learncpp.com/cpp-tutorial/a-strategy-for-debugging/>)
- 3.4 Basic debugging tactics (<https://www.learncpp.com/cpp-tutorial/basic-debugging-tactics/>)
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- 3.6 Using an integrated debugger: Stepping
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- 3.7 Using an integrated debugger: Running and breakpoints
(<https://www.learncpp.com/cpp-tutorial/using-an-integrated-debugger-running-and-breakpoints/>)

Using an integrated debugger: Watching variables

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Unsigned integers, and why to avoid them

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Introduction to if statements (<https://www.learncpp.com/cpp-tutorial/introduction-to-if-statements/>)

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Introduction to type conversion and `static_cast`

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N numeral systems (decimal, binary, hexadecimal, and octal)

5.3 [\(https://www.learncpp.com/cpp-tutorial/numeral-systems-decimal-binary-hexadecimal-and-octal/\)](https://www.learncpp.com/cpp-tutorial/numeral-systems-decimal-binary-hexadecimal-and-octal/)

5.4 [The as-if rule and compile-time optimization](https://www.learncpp.com/cpp-tutorial/the-as-if-rule-and-compile-time-optimization/) (<https://www.learncpp.com/cpp-tutorial/the-as-if-rule-and-compile-time-optimization/>)

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5.5 [Constant expressions](https://www.learncpp.com/cpp-tutorial/constant-expressions/) (<https://www.learncpp.com/cpp-tutorial/constant-expressions/>) Updated

5.6 [Constexpr variables](https://www.learncpp.com/cpp-tutorial/constexpr-variables/) (<https://www.learncpp.com/cpp-tutorial/constexpr-variables/>)

5.7 [Introduction to std::string](https://www.learncpp.com/cpp-tutorial/introduction-to-stdstring/) (<https://www.learncpp.com/cpp-tutorial/introduction-to-stdstring/>)

5.8 [Introduction to std::string_view](https://www.learncpp.com/cpp-tutorial/introduction-to-stdstring_view/) (https://www.learncpp.com/cpp-tutorial/introduction-to-stdstring_view/)

5.9 [std::string_view \(part 2\)](https://www.learncpp.com/cpp-tutorial/stdstring_view-part-2/) (https://www.learncpp.com/cpp-tutorial/stdstring_view-part-2/)

5.x [Chapter 5 summary and quiz](https://www.learncpp.com/cpp-tutorial/chapter-5-summary-and-quiz/) (<https://www.learncpp.com/cpp-tutorial/chapter-5-summary-and-quiz/>)

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6.2 [Arithmetic operators](https://www.learncpp.com/cpp-tutorial/arithmetic-operators/) (<https://www.learncpp.com/cpp-tutorial/arithmetic-operators/>)

6.3 [Remainder and Exponentiation](https://www.learncpp.com/cpp-tutorial/remainder-and-exponentiation/) (<https://www.learncpp.com/cpp-tutorial/remainder-and-exponentiation/>)

6.4 [Increment/decrement operators, and side effects](https://www.learncpp.com/cpp-tutorial/increment-decrement-operators-and-side-effects/) (<https://www.learncpp.com/cpp-tutorial/increment-decrement-operators-and-side-effects/>)

6.5 [The comma operator](https://www.learncpp.com/cpp-tutorial/the-comma-operator/) (<https://www.learncpp.com/cpp-tutorial/the-comma-operator/>)

6.6 [The conditional operator](https://www.learncpp.com/cpp-tutorial/the-conditional-operator/) (<https://www.learncpp.com/cpp-tutorial/the-conditional-operator/>)

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6.7 [Relational operators and floating point comparisons](https://www.learncpp.com/cpp-tutorial/relational-operators-and-floating-point-comparisons/) (<https://www.learncpp.com/cpp-tutorial/relational-operators-and-floating-point-comparisons/>)

6.8 [Logical operators](https://www.learncpp.com/cpp-tutorial/logical-operators/) (<https://www.learncpp.com/cpp-tutorial/logical-operators/>)

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Bit Manipulation (optional chapter)

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O.2 [Bitwise operators](https://www.learncpp.com/cpp-tutorial/bitwise-operators/) (<https://www.learncpp.com/cpp-tutorial/bitwise-operators/>)

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Scope, Duration, and Linkage

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User-defined namespaces and the scope resolution operator

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Why (non-const) global variables are evil

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Using declarations and using directives

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- 8.2 [If statements and blocks](https://www.learncpp.com/cpp-tutorial/if-statements-and-blocks/) (<https://www.learncpp.com/cpp-tutorial/if-statements-and-blocks/>)
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- 8.4 [Constexpr if statements](https://www.learncpp.com/cpp-tutorial/constexpr-if-statements/) (<https://www.learncpp.com/cpp-tutorial/constexpr-if-statements/>)
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- 8.6 [Switch fallthrough and scoping](https://www.learncpp.com/cpp-tutorial/switch-fallthrough-and-scoping/) (<https://www.learncpp.com/cpp-tutorial/switch-fallthrough-and-scoping/>)
- 8.7 [Goto statements](https://www.learncpp.com/cpp-tutorial/goto-statements/) (<https://www.learncpp.com/cpp-tutorial/goto-statements/>)
- 8.8 [Introduction to loops and while statements](https://www.learncpp.com/cpp-tutorial/introduction-to-loops-and-while-statements/) (<https://www.learncpp.com/cpp-tutorial/introduction-to-loops-and-while-statements/>)
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- 8.13 [Introduction to random number generation](https://www.learncpp.com/cpp-tutorial/introduction-to-random-number-generation/) (<https://www.learncpp.com/cpp-tutorial/introduction-to-random-number-generation/>)
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- 8.15 [Global random numbers \(Random.h\)](https://www.learncpp.com/cpp-tutorial/global-random-numbers-random-h/) (<https://www.learncpp.com/cpp-tutorial/global-random-numbers-random-h/>)
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- 9.2 [Code coverage](https://www.learncpp.com/cpp-tutorial/code-coverage/) (<https://www.learncpp.com/cpp-tutorial/code-coverage/>)
- 9.3 [Common semantic errors in C++](https://www.learncpp.com/cpp-tutorial/common-semantic-errors-in-c/) (<https://www.learncpp.com/cpp-tutorial/common-semantic-errors-in-c/>)
- 9.4 [Detecting and handling errors](https://www.learncpp.com/cpp-tutorial/detecting-and-handling-errors/) (<https://www.learncpp.com/cpp-tutorial/detecting-and-handling-errors/>)
- 9.5 [std::cin and handling invalid input](https://www.learncpp.com/cpp-tutorial/std-cin-and-handling-invalid-input/) (<https://www.learncpp.com/cpp-tutorial/stdcin-and-handling-invalid-input/>)
- 9.6 [Assert and static_assert](https://www.learncpp.com/cpp-tutorial/assert-and-static_assert/) (https://www.learncpp.com/cpp-tutorial/assert-and-static_assert/)

Type Conversion, Type Aliases, and Type Deduction

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- 10.1 [Implicit type conversion](https://www.learncpp.com/cpp-tutorial/implicit-type-conversion/) (<https://www.learncpp.com/cpp-tutorial/implicit-type-conversion/>)
- 10.2 [Floating-point and integral promotion](https://www.learncpp.com/cpp-tutorial/floating-point-and-integral-promotion/) (<https://www.learncpp.com/cpp-tutorial/floating-point-and-integral-promotion/>)
- 10.3 [Numeric conversions](https://www.learncpp.com/cpp-tutorial/numeric-conversions/) (<https://www.learncpp.com/cpp-tutorial/numeric-conversions/>)
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- 10.5 [Arithmetic conversions](https://www.learncpp.com/cpp-tutorial/arithmetic-conversions/) (<https://www.learncpp.com/cpp-tutorial/arithmetic-conversions/>)
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- 10.7 [Typedefs and type aliases](https://www.learncpp.com/cpp-tutorial/typedefs-and-type-aliases/) (<https://www.learncpp.com/cpp-tutorial/typedefs-and-type-aliases/>)
- 10.8 [Type deduction for objects using the auto keyword](https://www.learncpp.com/cpp-tutorial/type-deduction-for-objects-using-the-auto-keyword/) (<https://www.learncpp.com/cpp-tutorial/type-deduction-for-objects-using-the-auto-keyword/>)
- 10.9 [Type deduction for functions](https://www.learncpp.com/cpp-tutorial/type-deduction-for-functions/) (<https://www.learncpp.com/cpp-tutorial/type-deduction-for-functions/>)
- 10.x [Chapter 10 summary and quiz](https://www.learncpp.com/cpp-tutorial/chapter-10-summary-and-quiz/) (<https://www.learncpp.com/cpp-tutorial/chapter-10-summary-and-quiz/>)

Function Overloading and Function Templates

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- 11.1 [Introduction to function overloading](https://www.learncpp.com/cpp-tutorial/introduction-to-function-overloading/) (<https://www.learncpp.com/cpp-tutorial/introduction-to-function-overloading/>)
- 11.2 [Function overload differentiation](https://www.learncpp.com/cpp-tutorial/function-overload-differentiation/) (<https://www.learncpp.com/cpp-tutorial/function-overload-differentiation/>)
- 11.3 [Function overload resolution and ambiguous matches](https://www.learncpp.com/cpp-tutorial/function-overload-resolution-and-ambiguous-matches/) (<https://www.learncpp.com/cpp-tutorial/function-overload-resolution-and-ambiguous-matches/>)
- 11.4 [Deleting functions](https://www.learncpp.com/cpp-tutorial/deleting-functions/) (<https://www.learncpp.com/cpp-tutorial/deleting-functions/>)
- 11.5 [Default arguments](https://www.learncpp.com/cpp-tutorial/default-arguments/) (<https://www.learncpp.com/cpp-tutorial/default-arguments/>)
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- 11.8 [Function templates with multiple template types](https://www.learncpp.com/cpp-tutorial/function-templates-with-multiple-template-types/) (<https://www.learncpp.com/cpp-tutorial/function-templates-with-multiple-template-types/>)
- 11.9 [Non-type template parameters](https://www.learncpp.com/cpp-tutorial/non-type-template-parameters/) (<https://www.learncpp.com/cpp-tutorial/non-type-template-parameters/>)

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Constexpr functions

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F.1 [Constexpr functions](https://www.learncpp.com/cpp-tutorial/constexpr-functions/) (<https://www.learncpp.com/cpp-tutorial/constexpr-functions/>) Moved

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Compound Types: References and Pointers

Chapter 1

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Compound Types: Enums and Structs

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- 15.3 Nested types (member types)
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- 15.4 Introduction to destructors (<https://www.learncpp.com/cpp-tutorial/introduction-to-destructors/>)
- 15.5 Class templates with member functions
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- 15.6 Static member variables (<https://www.learncpp.com/cpp-tutorial/static-member-variables/>)
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- 15.9 Friend classes and friend member functions
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Dynamic arrays: `std::vector`

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- 16.1 Introduction to containers and arrays
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- 16.2 Introduction to `std::vector` and list constructors
(<https://www.learncpp.com/cpp-tutorial/introduction-to-stdvector-and-list-constructors/>)
- 16.3 `std::vector` and the unsigned length and subscript problem
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- 16.4 Passing `std::vector` (<https://www.learncpp.com/cpp-tutorial/passing-stdvector/>)
- 16.5 Returning `std::vector`, and an introduction to move semantics
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- 16.6 Arrays and loops (<https://www.learncpp.com/cpp-tutorial/arrays-and-loops/>)
- 16.7 Arrays, loops, and sign challenge solutions
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- 16.8 Range-based for loops (for-each)
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- 16.10 `std::vector` resizing and capacity
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Fixed-size arrays: `std::array` and C-style arrays

17.1 [Introduction to `std::array`](https://www.learncpp.com/cpp-tutorial/introduction-to-stdarray/) (<https://www.learncpp.com/cpp-tutorial/introduction-to-stdarray/>)

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Dynamic Allocation (under construction)

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19.4 [Pointers to pointers and dynamic multidimensional arrays](https://www.learncpp.com/cpp-tutorial/pointers-to-pointers/)
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Operator Overloading

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21.6 [Overloading unary operators +, -, and !](#)

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21.7 Overloading the comparison operators

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21.8 Overloading the increment and decrement operators

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21.10 Overloading the parenthesis operator

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21.11 Overloading typecasts [\(https://www.learncpp.com/cpp-tutorial/overloading-typecasts/\)](https://www.learncpp.com/cpp-tutorial/overloading-typecasts/)

21.12 Overloading the assignment operator

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Move Semantics and Smart Pointers

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22.1 Introduction to smart pointers and move semantics

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22.3 Move constructors and move assignment

[\(https://www.learncpp.com/cpp-tutorial/move-constructors-and-move-assignment/\)](https://www.learncpp.com/cpp-tutorial/move-constructors-and-move-assignment/)

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22.5 std::unique_ptr [\(https://www.learncpp.com/cpp-tutorial/stdunique_ptr/\)](https://www.learncpp.com/cpp-tutorial/stdunique_ptr/)

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22.7 Circular dependency issues with std::shared_ptr, and std::weak_ptr [\(https://www.learncpp.com/cpp-tutorial/circular-dependency-issues-with-stdshared_ptr-and-stdweak_ptr/\)](https://www.learncpp.com/cpp-tutorial/circular-dependency-issues-with-stdshared_ptr-and-stdweak_ptr/)

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Object Relationships

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- 23.4 Association (<https://www.learncpp.com/cpp-tutorial/association/>)
- 23.5 Dependencies (<https://www.learncpp.com/cpp-tutorial/dependencies/>)
- 23.6 Container classes (<https://www.learncpp.com/cpp-tutorial/container-classes/>)
- 23.7 `std::initializer_list` (https://www.learncpp.com/cpp-tutorial/stdinitializer_list/)
- 23.x Chapter 23 summary and quiz (<https://www.learncpp.com/cpp-tutorial/chapter-23-summary-and-quiz/>)

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- 24.1 Introduction to inheritance (<https://www.learncpp.com/cpp-tutorial/introduction-to-inheritance/>)
- 24.2 Basic inheritance in C++ (<https://www.learncpp.com/cpp-tutorial/basic-inheritance-in-c/>)
- 24.3 Order of construction of derived classes (<https://www.learncpp.com/cpp-tutorial/order-of-construction-of-derived-classes/>)
- 24.4 Constructors and initialization of derived classes (<https://www.learncpp.com/cpp-tutorial/constructors-and-initialization-of-derived-classes/>)
- 24.5 Inheritance and access specifiers (<https://www.learncpp.com/cpp-tutorial/inheritance-and-access-specifiers/>)
- 24.6 Adding new functionality to a derived class (<https://www.learncpp.com/cpp-tutorial/adding-new-functionality-to-a-derived-class/>)
- 24.7 Calling inherited functions and overriding behavior (<https://www.learncpp.com/cpp-tutorial/calling-inherited-functions-and-overriding-behavior/>)
- 24.8 Hiding inherited functionality (<https://www.learncpp.com/cpp-tutorial/hiding-inherited-functionality/>)
- 24.9 Multiple inheritance (<https://www.learncpp.com/cpp-tutorial/multiple-inheritance/>)
- 24.x Chapter 24 summary and quiz (<https://www.learncpp.com/cpp-tutorial/chapter-24-summary-and-quiz/>)

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- 25.1 Pointers and references to the base class of derived objects (<https://www.learncpp.com/cpp-tutorial/pointers-and-references-to-the-base-class-of-derived-objects/>)
- 25.2 Virtual functions and polymorphism (<https://www.learncpp.com/cpp-tutorial/virtual-functions/>)
- 25.3 The override and final specifiers, and covariant return types (<https://www.learncpp.com/cpp-tutorial/the-override-and-final-specifiers-and-covariant-return-types/>)
- 25.4 Virtual destructors, virtual assignment, and overriding virtualization (<https://www.learncpp.com/cpp-tutorial/virtual-destructors-virtual-assignment-and-overriding-virtualization/>)
- 25.5 Early binding and late binding (<https://www.learncpp.com/cpp-tutorial/early-binding-and-late-binding/>)
- 25.6 The virtual table (<https://www.learncpp.com/cpp-tutorial/the-virtual-table/>)

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25.9 Object slicing (<https://www.learncpp.com/cpp-tutorial/object-slicing/>)

25.10 Dynamic casting (<https://www.learncpp.com/cpp-tutorial/dynamic-casting/>)

25.11 Printing inherited classes using operator<<
(<https://www.learncpp.com/cpp-tutorial/printing-inherited-classes-using-operator/>)

25.x Chapter 25 summary and quiz
(<https://www.learncpp.com/cpp-tutorial/chapter-25-summary-and-quiz/>)

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26.1 Template classes (<https://www.learncpp.com/cpp-tutorial/template-classes/>)

26.2 Template non-type parameters
(<https://www.learncpp.com/cpp-tutorial/template-non-type-parameters/>)

26.3 Function template specialization
(<https://www.learncpp.com/cpp-tutorial/function-template-specialization/>)

26.4 Class template specialization
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26.5 Partial template specialization
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26.6 Partial template specialization for pointers
(<https://www.learncpp.com/cpp-tutorial/partial-template-specialization-for-pointers/>)

26.x Chapter 26 summary and quiz
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27.1 The need for exceptions (<https://www.learncpp.com/cpp-tutorial/the-need-for-exceptions/>)

27.2 Basic exception handling (<https://www.learncpp.com/cpp-tutorial/basic-exception-handling/>)

27.3 Exceptions, functions, and stack unwinding
(<https://www.learncpp.com/cpp-tutorial/exceptions-functions-and-stack-unwinding/>)

27.4 Uncaught exceptions and catch-all handlers
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27.5 Exceptions, classes, and inheritance
(<https://www.learncpp.com/cpp-tutorial/exceptions-classes-and-inheritance/>)

27.6 Rethrowing exceptions (<https://www.learncpp.com/cpp-tutorial/rethrowing-exceptions/>)

27.7 Function try blocks (<https://www.learncpp.com/cpp-tutorial/function-try-blocks/>)

- 27.8 [Exception dangers and downsides](https://www.learncpp.com/cpp-tutorial/exception-dangers-and-downsides/)
- 27.9 [Exception specifications and noexcept](https://www.learncpp.com/cpp-tutorial/exception-specifications-and-noexcept/)
- 27.10 [std::move_if_noexcept](https://www.learncpp.com/cpp-tutorial/stdmove_if_noexcept/)
- 27.x [Chapter 27 summary and quiz](https://www.learncpp.com/cpp-tutorial/chapter-27-summary-and-quiz/)

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- 28.1 [Input and output \(I/O\) streams](https://www.learncpp.com/cpp-tutorial/input-and-output-io-streams/)
- 28.2 [Input with istream](https://www.learncpp.com/cpp-tutorial/input-with-istream/)
- 28.3 [Output with ostream and ios](https://www.learncpp.com/cpp-tutorial/output-with-ostream-and-ios/)
- 28.4 [Stream classes for strings](https://www.learncpp.com/cpp-tutorial/stream-classes-for-strings/)
- 28.5 [Stream states and input validation](https://www.learncpp.com/cpp-tutorial/stream-states-and-input-validation/)
- 28.6 [Basic file I/O](https://www.learncpp.com/cpp-tutorial/basic-file-io/)
- 28.7 [Random file I/O](https://www.learncpp.com/cpp-tutorial/random-file-io/)

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Miscellaneous Subjects

Appendix

- A.1 [Static and dynamic libraries](https://www.learncpp.com/cpp-tutorial/a1-static-and-dynamic-libraries/)
- A.2 [Using libraries with Visual Studio](https://www.learncpp.com/cpp-tutorial/a2-using-libraries-with-visual-studio-2005-express/)
- A.3 [Using libraries with Code::Blocks](https://www.learncpp.com/cpp-tutorial/a3-using-libraries-with-codeblocks/)
- A.4 [C++ FAQ](https://www.learncpp.com/cpp-tutorial/cpp-faq/)

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Appendix

- B.1 [Introduction to C++11](https://www.learncpp.com/cpp-tutorial/introduction-to-c11/)
- B.2 [Introduction to C++14](https://www.learncpp.com/cpp-tutorial/introduction-to-c14/)
- B.3 [Introduction to C++17](https://www.learncpp.com/cpp-tutorial/introduction-to-c17/)
- B.4 [Introduction to C++20](https://www.learncpp.com/cpp-tutorial/introduction-to-c20/)
- B.5 [Introduction to C++23](https://www.learncpp.com/cpp-tutorial/introduction-to-c23/)

C.1 [The end?](https://www.learncpp.com/cpp-tutorial/appendix-c-the-end/) (<https://www.learncpp.com/cpp-tutorial/appendix-c-the-end/>)

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21.2

[STL containers overview](https://www.learncpp.com/cpp-tutorial/stl-containers-overview/) (<https://www.learncpp.com/cpp-tutorial/stl-containers-overview/>)

21.3 [STL iterators overview](https://www.learncpp.com/cpp-tutorial/stl-iterators-overview/) (<https://www.learncpp.com/cpp-tutorial/stl-iterators-overview/>)

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22.1 [std::string and std::wstring](https://www.learncpp.com/cpp-tutorial/stdstring-and-stdwstring/) (<https://www.learncpp.com/cpp-tutorial/stdstring-and-stdwstring/>)

22.2 [std::string construction and destruction](https://www.learncpp.com/cpp-tutorial/stdstring-construction-and-destruction/)
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22.3 [std::string length and capacity](https://www.learncpp.com/cpp-tutorial/stdstring-length-and-capacity/)
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(<https://www.learncpp.com/cpp-tutorial/stdstring-assignment-and-swapping/>)

22.6 [std::string appending](https://www.learncpp.com/cpp-tutorial/stdstring-appending/) (<https://www.learncpp.com/cpp-tutorial/stdstring-appending/>)

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